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Abstract	This work aims to compare Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) as fitness functions in a genetic algorithm used to optimize the hyperparameters of an Echo State Network (ESN) with five probabilistic distributions. We have used financial time series data extracted from the Brazilian Stock Market. The ESN proved to be excellent for predicting time series. However, using MAE and RMSE within the genetic algorithm was ineffective since the results did not differ significantly despite the asymmetrical data distribution. Two of the five distributions stood out, Uniform and Student's t, showing low variability both with the use of RMSE and MAE, where we graphically observed that Uniform presented better prediction using the MAE fitness.
Keywords (separated by '-')	Stock price forecasting - Financial Time Series - ESN

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Comparing MAE and RMSE as Fitness of Genetic Algorithm for Optimizing Echo State Network Hyperparameters with Different Probabilistic Distributions

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Abstract. This work aims to compare Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) as fitness functions in a genetic algorithm used to optimize the hyperparameters of an Echo State Network (ESN) with five probabilistic distributions. We have used financial time series data extracted from the Brazilian Stock Market. The ESN proved to be excellent for predicting time series. However, using MAE and RMSE within the genetic algorithm was ineffective since the results did not differ significantly despite the asymmetrical data distribution. Two of the five distributions stood out, Uniform and Student's t, showing low variability both with the use of RMSE and MAE, where we graphically observed that Uniform presented better prediction using the MAE fitness.

Keywords: Stock price forecasting · Financial Time Series · ESN

1 Introduction

Time Series (TS) forecasting involves mathematical models and statistical techniques, which can obtain patterns in data and use them to make predictions. Therefore, Echo State Network (ESN) is quite viable, as it presents more volatile training. However, there is an optimization problem in its hyperparameters, as there are certain degrees of randomness in its architecture, causing variation problems [16]. In order to optimize the hyperparameters, we use MAE and RMSE as fitness within Genetic Algorithm (GA).

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With modernity, the financial market has become increasingly complex and volatile, characterized as a system where purchases and sales of assets and other financial instruments occur. The research field in this area is critical and challenging, as it demands much effort to work with TS since they have big data.

Stock forecasting significantly affects companies, individuals, and countries, with the ability to determine the direction they may go in the future. When dealing with countries, stock forecasting can help control economic risks and stabilize the financial market. Data analysis can help companies avoid future crises in their businesses and the processing of their data. Furthermore, investors can use stock closing data to plan for future asset purchases or protect themselves from market risks [22].

The frequent increase in the study of regressions in the field of statistics aimed at forecasting stock data generated a deepening in the use of error metrics together with the Genetic Algorithm, where they function as fitness to optimize the hyperparameters of the ESN [10] seeking more incredible accuracy. Despite ESN not requiring almost any training, it has two disadvantages: the number of hyperparameters to be adjusted [12] and the necessity to have the input data standardized, as all Artificial Neural Networks (ANN) [14,21]. According to [7], GA has some advantages: they do not require assuming any specific statistical distribution of the given data; they do not require the specification of the model form; and they deal effectively with non-stationary data. These characteristics help to deal with the first ESN disadvantage.

One way to determine the accuracy of a regression model is to calculate the difference between what is observed in the TS and what is forecasted. The error can be calculated in various ways because it can not simply add up the errors since there are negative and positive values that end up canceling each other out. As a result, several strategies have been adopted to avoid cancellation. Some examples are the Mean Absolute Error (MAE), where the average error is calculated as a module, and the Mean Squared Error (RMSE), where the values are squared, their average calculated, and the square root is applied.

This study proposes comparing different error measures as fitness within a genetic algorithm that contributes to optimizing the hyperparameters fitted by an ESN using different probabilistic distributions. We have used five probability distributions in this study: normal, uniform, Student's t, generalized error distribution (GED), and Asymmetric Student's t, to analyze which would be the most accurate.

This article has been organized as follows. Section 2 presents the theoretical foundation and a literature review on the Echo State Network, Genetic Algorithm, and probabilistic distributions. Section 3 details material and methods. Section 4 contains and discusses the results obtained. Finally, Sect. 5 draws the conclusion and some future work.

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2 Theoretical Foundation and Literature Review

Stock market forecasting is challenging for generating non-stationary TS and has randomness. There are two hypotheses: the Efficient Market Hypothesis (EMH) asserts that price changes in the capital market can occur independently, although several empirical studies have demonstrated that stock market predictions are possible. EMH can be divided into three types (weak, semi-strong, and strong) according to the level of reflection of market information [6].

Previous studies have employed statistical and/or machine learning techniques to forecast future values [2]. We highlight some [18] compared three neural network models: time delay, recurrent and probabilistic neural networks, and employed conjugate gradient and multi-stream extended Kalman filter training methods for Time Delay Neural Network (TDNN) and Recurrent Neural Network (RNN) for predicting stock trends. RNN showed the best performance among the models. [5] Used a neural network, a TS fuzzy system, and a hierarchical fuzzy system to verify the effectiveness of the hybrid model. It was found that the use of each one proved to be viable, concluding that there was no better.

The ESN, proposed by [10], is an RNN model. It uses reservoir calculation, where the neurons are fixed, receiving random weights that will be trained at the output. His study was only theoretical, but in 2004, it was applied in areas related to wireless communication. As an alternative to unstable and more complex RNN, the ESN has been increasingly improved to simplify its training. Thus, it has a shorter and faster cycle, in addition to using many neurons, dynamizing the network. Furthermore, it presents significant contributions in the field of TS prediction, with applicability in object tracking [22].

The GAs are metaheuristic and stochastic optimization algorithms inspired by the process of natural evolution [9]. They are used to find near-optimal solutions to optimization problems with large search spaces. The process of GA includes operators that imitate natural genetic and evolutionary principles, such as crossover and mutation. The prominent feature of GA is the population of "chromosomes" and the possibility of reaching a local optimum using them [1].

In this context, we have the normal distribution, being the most important and used in statistical circles, which is characterized as continuous and symmetrical, representing the behavior of a natural phenomenon randomly. In addition to differing in their mean and standard deviation values, its probability density function can be as follow: $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$, where: μ is the mean of the distribution, σ^2 is the variance of the distribution,x is the aleatory variable. [8]

The uniform distribution is a fundamental statistical probability distribution characterized by assigning the same probability to all possible outcomes within a defined range. Consisting of grouping similar data points [8]. The probability density function for a continuous uniform distribution is given by: $f(x; a, b) = \frac{1}{(b\text{-}a)}$ [4], where x is a random variable.

The Student's t distribution is presented as a fundamental tool in inferential statistics. This distribution is used to estimate the mean of a normally

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distributed population in situations where the sample size is small and the population standard deviation is unknown, its probability density function can be as [26]: $S(\boldsymbol{v}, \boldsymbol{K}, k) = \frac{\Gamma\left(\frac{d+k}{2}\right)}{\Gamma\left(\frac{k}{2}\right)} \cdot (k\pi)^{-\frac{d}{2}} \cdot |\boldsymbol{K}|^{-\frac{1}{2}} \cdot \left(1 + \frac{1}{k}(\boldsymbol{v} - \boldsymbol{l})^T \boldsymbol{K}(\boldsymbol{v} - \boldsymbol{l})\right)^{-\frac{d+k}{2}}$ where: Γ is the Gamma function; ν is the number of degrees; d is a variable; $\boldsymbol{K}$ is a precision matrix.

The GED distribution is a generalization of the normal distribution, proposed by [15], allowing greater flexibility in modeling data with tails that contain larger or smaller weights, tending towards one side of the training, its probability density function can be as follow [13]: $f(x|\mu, \sigma, \beta) = \frac{\beta}{2\sigma\Gamma\left(\frac{1}{\beta}\right)} \exp\left(-\left|\frac{x-\mu}{\sigma}\right|^{\beta}\right)$ where: μ is the average, σ is the scale parameter, β is the shape parameter that controls the kurtosis of the distribution, Γ is the Gamma function.

The asymmetric Student-t distribution is a form contrary to the conventional Student-t, where it is continuous and unimodal with a focus on its center, differentiating at any point at least once [25].

3 Material and Methods

The study of TS is significant for learning how the stock market works, the techniques used to make forecasts, and how to use this knowledge. To do this, we obtained the data set of the stock's closing prices under study, processed it, and set up our experiments using the R program language, in which some libraries were parameterized to improve the results. GA was used to optimize the ESN hyperparameters. In our case, each chromosome acts as a potential solution of a hyperparameter, and it is expressed in the form of binary strings. These chromosomes are generated randomly, and the one that provides the better solution gets more chance to reproduce [1].

In order to optimize, it was necessary to use the GA library, with the following arguments: type = "binary" (binary type), nBits = 55 (chromosome size), popSize = 10 (number of individuals in the population), population = gabin_Population (population for binary chromosomes), pcrossover = 0.8 (population crossover of 80%), crossover = gabin_spCrossover (crossover between binary chromosomes), selection = gabin_tourSelection (generates a new population of individuals from the current population probabilistically, according to individual aptitudes), elitism = 1 (the best individual only), keepBest = T (keeps the best individual), mutation = gabin_raMutation (population mutation), mutation = 0.1 (mutation of 10%), maxiter = 10000 (maximum number of epochs), run = 3500 (maximum number of unimproved epochs).

The genetic algorithm's fitness calculation used the MAE and RMSE as criteria for evaluating the model, assigning 40% of the training MAE and 60% of the validation MAE to optimization, also serving for the RMSE fitness.

This section outlines the methodology used in the study, describing since as data was obtained until the metrics used to evaluate the performance of the Echo State Network (ESN) with hyperparameters optimized by the Genetic Algorithm (GA) with fitness MAE (Mean Absolute Error) and with fitness RMSE (Root

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Mean Squared Error), and different probabilistic distributions. Figure 1 shows the steps to make this project.

This project is based on four stages, as seen in Fig. 1. The first is to obtain the data. The second is to treat data. In the third, we used two techniques that worked together, ESN and GA, for optimization, in which we used different probabilistic distributions in ESN and included error metrics as fitness in GA. In the fourth step, we evaluate the error forecast using some metrics.

3.1 Obtaining the Data

This software and datasets were obtained free of charge, and data on daily stock prices were obtained from the data set BovDB[1] [3]. So, we used log returns data calculated from this TS (TS) relating to the closing price of a stock. In this way, these data were divided into training and validation.

Data on intraday stock prices were obtained using the Metatrader software, version 5.00, where intraday closing prices were obtained every 30 min and used to calculate daily variances. So, we used these data to test and to ensure greater efficiency and generalization of the model.

3.2 Data Treatment and Setup

Once the data has been obtained, the tools and models used to evaluate the TS were defined. To do this, graphs and tables were generated, including training, validation, test results, and the hyperparameters we seek to optimize.

Data treatment is based on the calculation of returns and variance, as shown: $R_t = \frac{p_t - p_{t-1}}{p_{t-1}}$ and $s^2 = \frac{\sum_{i=1}^{n}(p_i - \overline{p})^2}{(n-1)}$. First, the TS return values are used with the ESN, which will receive and process the dataset. Then, the daily variance values are used to finalize the results, where tests are carried out to see if the values have been well optimized.

The data was divided into training, validation, and testing. Initially, a total size of 5571 was used, divided into the three mentioned: adjusting training to receive 3335, validation with 1118, and testing with 1118. In this way, the values were divided, respectively, into 60% from the beginning of the TS, 20% from the sequence of the TS, and 20% from the end of the TS, in addition to using 40% of the data as an optimization weight for training errors and 60% as a weight for validation errors. With this, the model will use more memory for training and equal between testing and validation; this format is a classic option for ESNs as a static split in their initialization.

To carry out the experiments, it was necessary to configure the parameters used by the GA within the ESN, as follows: binary code with chromosome size of 55 bits; population of 10 individuals; population crossover of 80%; maximum limit of 10000 epochs; maximum unimproved epochs of 3500; population elitism of 10%; population mutation of 10%; and always keeping the best individual.

[1] Download available at https://github.com/Ginfofinance/BovDBrepository.

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We divide GA processing into six steps: initialization, fitness calculation, termination condition check, selection, crossover, and mutation. In the initialization stage, the generated chromosomes are selected arbitrarily and then undergo a fitness process calculated by the predefined metric. The fitness function is a method used to measure the performance of this chromosome. Optimization using a genetic algorithm requires the definition of fitness. Through this function, only the solutions with the best performance will be kept to repeat the process, characterizing the termination condition check stage. The superior chromosomes are reused in the selection process to reproduce better ones. This way, new solutions, such as the crossover stage, will be created from existing ones. Finally, in the mutation phase, one of the chromosomes is chosen to change a random bit. The objective of the process is to increase the set of solutions [6].

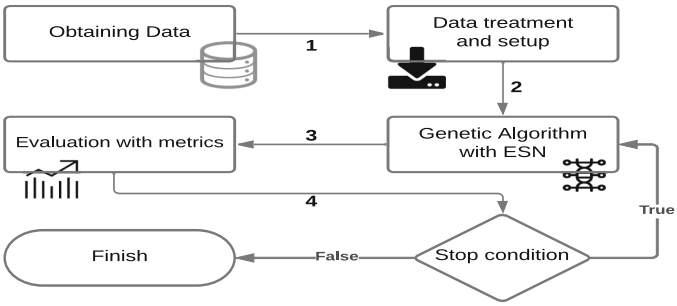


Fig. 1. Methodology of ESN with GA for tuning hyperparameters.

Figure 1 shows all the processing GA carries out with the parameters passed in. Initially, a new chromosome is generated through selection, crossover, and mulling, finally arriving at the stop condition. If the condition is not met, the selection, crossover, and mulling processes are repeated, creating a superior chromosome. In this study, the fitness evolution used are mean absolute error (MAE) and root-mean-squared error (RMSE).

This configuration in the GA refines the hyperparameters returned by the ESN in the statistical distributions used. The ESN hyperparameters used in this work are five: Leaking rate (α), Spectral radius (sr), Disregarded Initial Values (DIV), Reservoir size, and Regularization coefficient. Respectively, the leaking rate controls the speed with which the reservoir state is updated in response to new input weights, which are memories. The second is responsible for the stability of the neural network; if the spectral radius of the weights $\mathbf{W^r}$ within the reservoir is less than 1, ESN will have an echo state property. The reservoir size refers to the modeling capacity and performance of the network. A larger size means a more accurate stock forecast, but there will be an overfitting problem, while too small a size can lead to underfitting. Finally, the regularization coefficient helps to avoid overfitting the training data, balancing training accuracy and network compactness [24].

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In this study, the hyperparameters tuned to ESN, with the optimization of the GA, were used for testing and validation with five different distributions: Normal, Student's T, Uniform, GED, and Asymmetric Student's t. These were used to find out which of them best delivers values closest to 0, since the lower the value, greater the accuracy. To do this, we used the closing returns of the ITSA4 stock and calculated the best error using MAE and RMSE.

3.3 Evaluation with Metrics

We used the stock returns within the GA. With this, we compared the data calculated by ESN, observing the accuracy and evolution of performance based on the forecast samples and natural samples, using the mean absolute error (MAE) $[\frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|]$ and root-mean-square error (RMSE) $[\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}]$ metrics [11]. Where y_i and y_i are the actual and predicted values, respectively, up to the time i and n is the total number of predicted values in the experiment.

We executed the experiments in R programming language with RStudio statistical software. The libraries used for the study are useful and reliable for this work. The libraries used in RStudio were GA [20] to execute the Genetic Algorithm, PerformanceAnalytics [17] to calculate asymmetry and kurtosis, and ggplot2 [23], to plot graphics.

4 Experiment Results and Discussion

ITSA4 from January 2nd, 2001 to July 30th, 2023. To obtain the returns, we used the BovDB [3] data set with existing closing stock prices during the mentioned period and calculated them as cited before. The data was divided into training, validation, and testing, following the machine learning standard: 60% for training, 20% for validation, and 20% for testing. In the table above, we compared the ESN hyperparameters within each distribution using the RMSE as fitness to find out which would best return the expected result, i.e., with greater precision, seeking to get closer to zero.

Table 1 shows the ESN hyperparameters tuned by GA using the RMSE error metric as fitness along with the probabilistic distributions, where the Leaking rate, sr, DIV, Reservoir size, and Regularization coefficient were analyzed.

In it, we compared the ESN hyperparameters within each distribution using the RMSE as fitness to find out which would best return the expected result, that is, with greater precision, seeking to get closer to zero. Looking at them, we noted that by using the RMSE for tuning, some values were close, and we noticed that uniform showed higher values while Student's t returned lower values.

In Table 2, we made the same comparison but using the MAE metric, and it is possible to see a decrease in values in the uniform distribution compared to the GED. However, analyzing the variability in general, we saw that it had an increase in values, and while using the RMSE, we had lower results.

Figure 2 graphically shows the attempt to predict the target TS characterized in green and the predicted series in black, where the two distributions are using

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Table 1. ESN's hyperparameters using fitness RMSE.

Distribution	Epochs	Leaking rate (α)	Spectral radius (sr)	DIV	Reservoir size	Regularization coefficient
Normal	7.035	0.9245	0.2372	97	16	3.77E-05
	4.576	0.9348	0.3721	5	19	2.30E-05
	7.045	0.8197	0.1776	118	25	7.38E-05
Student's T	8.534	**0.7559**	0.7820	125	**15**	**2.75E-05**
	3.534	0.8632	**0.2286**	**99**	16	8.19E-05
	8.294	0.8624	0.4698	109	18	3.62E-05
Uniform	6.710	0.8660	0.4328	101	14	8.64E-05
	8.445	0.9670	**0.2512**	**21**	**15**	7.84E-05
	4.943	0.5148	0.8033	39	23	**1.43E-05**
GED	7.804	0.9511	0.4108	98	7	2.03E-05
	5.270	0.7538	0.6595	48	6	4.05E-05
	6.913	0.9116	0.5740	96	19	2.36E-05
Asymmetric Student's T	5.812	0.9439	0.5175	66	31	6.04E-05
	7.331	0.8871	0.1840	95	32	1.00E-05
	4.808	0.8664	0.7923	33	22	9.90E-05
Variability						
normal	2.469	0.1151	0.1945	113	9	5.08E-05
Student's t	5.000	0.1073	0.5534	26	3	5.44E-05
uniform	3.502	0.2593	0.0732	97	4	7.21E-05
GED	2.534	0.1973	0.2487	50	12	2.02E-05
Asymmetric Student's T	2.523	0.0775	0.6083	62	10	8.90E-05

Table 2. ESN's hyperparameters using fitness MAE.

Distribution	Epochs	Leaking rate (α)	Spectral radius (sr)	DIV	Reservoir size	Regularization coefficient
Normal	6,493	0,5312	0,7396	30	29	9,90E-05
	3,741	0,5475	0,9233	83	30	9,90E-05
	4,630	0,5428	0,9231	24	27	1,74E-05
Student's t	5,870	0,9539	0,1401	26	31	1,94E-05
	8,143	0,9195	0,6965	57	9	9,06E-05
	3,784	0,7343	0,5192	11	27	1,76E-05
uniform	7,582	0,5382	0,9174	85	31	5,91E-05
	4,179	0,6920	0,9730	29	27	9,90E-05
	4,048	0,7975	0,8998	126	28	1,36E-05
GED	5,361	0,6485	0,3237	61	12	8,23E-05
	5,008	0,8844	0,5590	27	10	6,64E-05
	4,626	0,3936	0,9479	128	3	3,46E-05
Asymmetric Student's t	8,112	0,5262	0,9912	10	33	1,94E-05
	3,957	0,9182	0,4855	57	17	1,84E-05
	7,836	0,8560	0,2920	91	20	3,54E-05
Variability						
Normal	2,752	0,0163	0,1897	59	2	8,16E-05
Student's T	4,359	0,2196	0,5564	46	22	7,30E-05
uniform	3,534	0,2593	0,0732	97	4	8,54E-05
GED	735	0,4908	0,6242	101	9	4,77E-05
Asymmetric Student's t	4,155	0,3920	0,6992	81	16	1,70E-05

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Table 3. Comparison with fitness MAE

Distribution	Epochs	MAE			RMSE		
		Training (%)	Validate (%)	Test	Training (%)	Validate (%)	Test
Student t	5870	0.01640	0.01497	0.00195	0.02260	0.02050	0.00425
	8143	0.01631	0.01506	0.00246	0.02220	0.02050	0.00473
	3784	0.01639	0.01501	0.00228	0.02240	0.02050	0.00457
Mean	5932.33	0.01637	0.01503	0.00223	0.02240	0.02058	0.00452
Asymmetric Student t	8112	0.01626	0.01499	0.00193	0.02215	0.02052	0.00430
	3957	0.01644	0.01496	0.00213	0.02258	0.02054	0.00435
	7836	0.01645	0.01497	0.00204	0.02261	0.02056	0.00430
Mean	6635	0.01638	**0.01497**	0.00203	0.02244	0.02215	0.00431
uniform	7582	0.01626	0.01512	0.00192	0.02198	0.02063	0.00434
	4179	0.01626	0.01510	0.00183	0.02207	0.02055	0.00428
	4048	0.01629	0.01509	0.00176	0.02208	0.02058	0.00395
Mean	5269.66	0.01627	0.01510	**0.00184**	**0.02198**	0.02058	**0.00419**
Normal	6493	0.01625	0.01505	0.00214	0.02205	0.02052	0.00429
	3741	0.01625	0.01506	0.00180	0.02204	0.02053	0.00434
	4630	0.01628	0.01513	0.00191	0.02209	0.02061	0.00418
Mean	4954.66	**0.01626**	0.01508	0.00195	0.02206	**0.02055**	0.00427
GED	5361	0.01651	0.01501	0.00219	0.02275	0.02065	0.00446
	5008	0.01649	0.01501	0.00186	0.02272	0.02063	0.00426
	4626	0.01650	0.01500	0.00226	0.02275	0.02063	0.00451
Mean	4998.33	0.01650	0.01501	0.00210	0.02275	0.02063	0.00441
Variability		Training (%)	Validate (%)	Test	Training (%)	Validate (%)	Test
Student t	4359	0.00012	0.00009	0.00051	0.00033	0.00030	0.00048
Asymmetric Student t	4155	0.00019	0.00003	0.00020	0.00046	0.00040	**0.00005**
Uniform	3534	0.00019	0.00003	**0.00016**	**0.00010**	0.00080	0.00039
Normal	2752	0.00003	0.00008	0.00034	0.00050	0.00090	0.00016
GED	735	**0.00002**	**0.00001**	0.00040	0.00030	**0.00020**	0.00025

the MAE fitness, thus comparing the accuracy of the two. We can see from this that the Uniform distribution was significantly more accurate than the Student's t. The red circles shows this.

The increase in the forecast series between the years 2019 and 2023 was more significant in the uniform because it had slightly smaller errors than the Student's t. There was also influence from the properties of each distribution.

Table 3 shows the errors obtained with the ESN in the training, validation, and testing stages. It can be seen that when we train and validate with different distributions, the errors hardly change between the distributions. Considering the fitness MAE and the mean values of the tests, the best performance was with uniform distribution. Considering the variability in the tests better results were obtained by uniform with MAE and by Asymmetric Student t with RMSE.

This table also shows the variability calculated for each distribution, and it is worth noting that the two, uniform and normal, had alternating MAE and RMSE over the test period. Besides analyzing test results and less variability,

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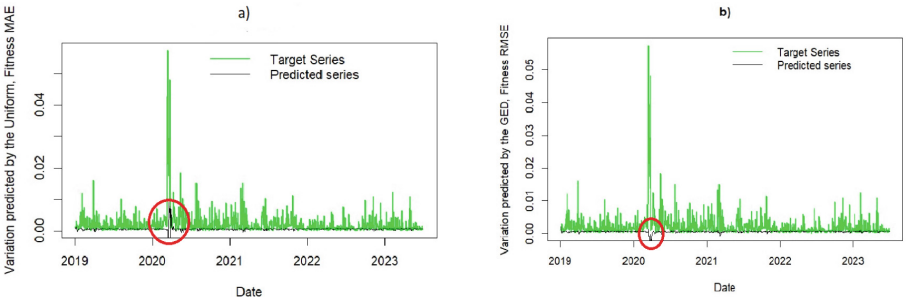


Fig. 2. Comparative Uniform and GED MAE(a) and RMSE(b).

Table 4. Comparison with fitness RMSE

Distribution	Epochs	MAE			RMSE		
		Training (%)	Validate (%)	Test	Training (%)	Validate (%)	Test
Student t	8534	0.01642	0.01499	0.00173	0.02251	0.02058	0.00415
	3534	0.01644	0.01498	0.00204	0.02261	0.02057	0.00429
	8294	0.01645	0.01497	0.00187	0.02262	0.02057	0.00427
Mean	6787.33	0.01644	**0.01498**	0.00188	0.02258	**0.02057**	**0.00424**
Asymmetric Student t	5812	0.01642	0.01498	0.00201	0.02248	0.02056	0.00430
	7331	0.01643	0.01497	0.00212	0.02256	0.02056	0.00432
	4808	0.01640	0.01499	0.00221	0.02250	0.02058	0.00437
Mean	5983.66	**0.01642**	**0.01498**	0.00211	**0.02251**	**0.02057**	0.00433
Uniform	6710	0.01648	0.01500	0.00184	0.02268	0.02060	0.00419
	8445	0.01647	0.01498	0.00184	0.02267	0.02058	0.00415
	4943	0.01642	0.01503	0.00202	0.02250	0.02063	0.00460
Mean	6699.33	0.01646	0.01500	0.00190	0.02262	0.02060	0.00431
Normal	7035	0.01646	0.01499	0.00211	0.02262	0.02058	0.00432
	4576	0.01645	0.01498	0.00214	0.02258	0.02056	0.00433
	7045	0.01649	0.01498	0.00198	0.02267	0.02056	0.00427
Mean	6218.66	0.01647	**0.01498**	0.00208	0.02262	**0.02057**	0.00431
GED	7804	0.01649	0.01502	0.00186	0.02273	0.02063	0.00425
	5270	0.01648	0.01502	0.00186	0.02271	0.02063	0.00427
	6913	0.01648	0.01502	0.00186	0.02272	0.02063	0.00426
Mean	6662.33	0.01648	0.01502	**0.00186**	0.02272	0.02063	0.00426
Variability		Training (%)	Validate (%)	Test	Training (%)	Validate (%)	Test
Student t	5000	0.00003	0.00002	0.00031	**0.00001**	0.00001	**0.00002**
Asymmetric Student t	2523	0.00003	0.00002	0.00020	0.00008	0.00002	0.00007
Uniform	3502	0.00008	0.00005	0.00018	0.00002	0.00005	0.00045
Normal	2469	0.00004	0.00001	0.00016	0.00009	0.00002	0.00006
GED	891	**0.00001**	**0.00000**	**0.00000**	0.00002	**0.00000**	**0.00002**

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we consider that using MAE as fitness, the best results were with uniform distribution for MAE and normal distribution for RMSE.

In Table 4, the errors obtained using the ESN are also shown, using the same steps as the machine training. However, a comparison was made using the fitness RMSE for the TS; in the table, we see that when it comes to training and validation, there are no significant variations, but when it comes to the tests, it is possible to see that GED obtained the better results with MAE, considering the mean in the tests and Student t the better results with RMSE. Considering the variability in the tests, the best results were obtained by GED in MAE and by GED and Student t in RMSE.

A parallel study focused on the methodology used in applying TS data mining techniques to calculate exchange rates of US dollars against Indian rupees. Furthermore, he used regression and language as in this work, also applying his considerations on the MAE and RMSE analyses. Therefore, it was concluded that RMSE applies greater value to large errors, generating outliers, while MAE does not have major influences when dealing with large errors [19].

5 Final Considerations

In this work, we used MAE and RMSE as the fitness of a genetic algorithm to optimize the hyperparameters of an ESN and compared the test results obtained.

The use of ESN showed that it is an excellent machine learning technique for forecasting TS, as the validation, training, and testing stages had very close values. We observed that the accuracy measured by both MAE and RMSE, the minimum average errors were less than 1% in all cases.

The change of MAE by RMSE within GA was not very effective since the data obtained did not differ significantly. Therefore, either MAE or RMSE can be used. However, two of the five distributions had the lowest test and variability values, namely uniform and GED. In this case, using GED with RMSE fitness had the best MAE and RMSE test results.

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Chapter 26

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